Computer Vision Practical Report:
GAUSSIAN BLURRED IMAGE

1. AIM

To read an input color image using OpenCV, apply Gaussian Blur to the image, display both the original image and Gaussian-blurred image side-by-side on the same page using Matplotlib, and save the blurred image to disk.

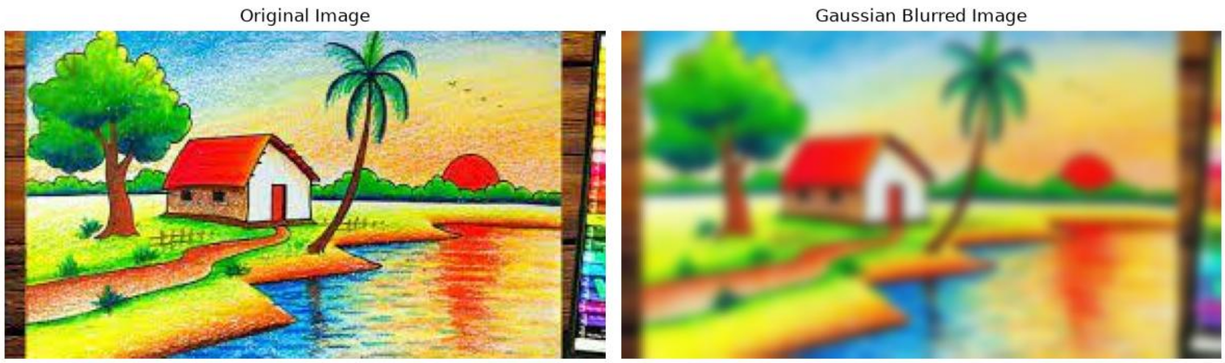
2. PROCEDURE

- Load the input image using cv2.imread('E2-IMG.png').
- Convert the image from BGR to RGB using cv2.cvtColor().
- Apply Gaussian Blur using cv2.GaussianBlur() with a (5,5) kernel.
- Create a Matplotlib figure of size 10x5.
- Display the original image and blurred image side-by-side using plt.subplot().
- Set titles as 'Original Image' and 'Gaussian Blurred Image'.
- Disable axes using plt.axis('off').
- Display the images using plt.show().
- Save the blurred image as **gaussian_blurred_image.png** using cv2.imwrite().

3. PROGRAM CODE

```
import cv2
import matplotlib.pyplot as plt

img_bgr = cv2.imread('E2-IMG.png')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

blurred_img = cv2.GaussianBlur(img_rgb, (15, 15), 0)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(blurred_img)
plt.title('Gaussian Blurred Image')
plt.axis('off')

plt.tight_layout()
plt.show()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image 'E2-IMG.png' was loaded, converted into a gaussian blur image.